Oracle Database Free

Description

Oracle AI Database 26ai Free Container Image Documentation

Oracle AI Database 26ai Free is the free edition of the industry-leading database. The Oracle AI Database 26ai Free Container Image contains Oracle AI Database 26ai Free based on an Oracle Linux 8 base image.

Two flavors of the image are supported:

1. The `Full` image: supports all the database features provided by Oracle AI Database 26ai Free.
2. The `Lite` image: smaller image size with a stripped-down installation of the database.

The `Lite` image has a smaller storage footprint than the `Full` image (~80% image size reduction) and a substantial improvement in image pull time. This image is useful in CI/CD scenarios and for simpler use cases where advanced database features are not required.

For more information on Oracle AI Database 26ai Free, see: https://oracle.com/database/free

Using the Full image

Starting an Oracle AI Database 26ai Free container

The Oracle AI Database 26ai Free Container Image contains a pre-built database, so the **startup time is very fast**. Fast startup can be helpful in CI/CD scenarios. To start an Oracle AI Database Free

Short URL for Repo

https://container-registry.oracle.com/ords/ocr/ba/database/free

Other Open Source Licenses

Pull Command for Latest

```
docker pull
```

```
podman run -d --name <oracle-db> container-registry.oracle.com/database/free:latest
```

When the container is started, a random password is generated for the `SYS, SYSTEM and PDBADMIN` users. This is termed the default password.

Note:

- Throughout this document, words enclosed within angle brackets `<` `>` indicate variables in code lines.
- To change the default password, see: "*Changing the Default Password for SYS User*"
- To learn about advanced use cases, see: "*Custom Configurations*"
- This document uses `podman` as the prescribed container runtime, but using contemporary commands is also anticipated to work.
- The Oracle Enterprise Manager Database Express (OEM DB Express) is no longer supported with Oracle AI Database 26ai Free. Please use SQL Developer instead.

The Oracle AI Database is ready to use when the `STATUS` field shows `(healthy)` in the `podman ps` output.

Custom Configurations

To facilitate custom configurations, the Oracle AI Database container provides configuration parameters that you can use when starting the container. For example, this is the detailed `podman run` command supporting all custom configurations:

```
podman run --name <container name=""> \
-P | -p <host port="">:1521 \
-e ORACLE_PWD=<your database="" passwords=""> \
-e ORACLE_CHARACTERSET=<your character="" set=""> \
-e ENABLE_ARCHIVELOG=true \
-e ENABLE_FORCE_LOGGING=true \
-v [<host mount="" point="">:]/opt/oracle/oradata \
container-registry.oracle.com/database/free:latest

Parameters:
   --name:        The name of the container (default: auto generated)
   -P | -p:       The port mapping of the host port to the container port.
                  Only one port is exposed: 1521 (Oracle Listener)
   -e ORACLE_PWD: The Oracle AI Database SYS, SYSTEM and PDB_ADMIN password (default: auto g
```

```
  -e ENABLE_ARCHIVELOG:
                  To enable archive log mode when creating the database (default: true)
  -e ENABLE_FORCE_LOGGING:
                  To enable force logging mode when creating the database (default: true)
  -v /opt/oracle/oradata
                  The data volume to use for the database.
                  Has to be writable by the Unix "oracle" (uid: 54321) user inside the conta
                  If omitted the database will not be persisted over container recreation.
  -v /opt/oracle/scripts/startup
                  Optional: A volume with custom scripts to be run after database startup.
                  For further details see the "Running scripts after setup and on startup" s
  -v /opt/oracle/scripts/setup
                  Optional: A volume with custom scripts to be run after database setup.
                  For further details see the "Running scripts after setup and on startup" s
```

The supported configuration options are:

- **ORACLE_CHARACTERSET:**
  This parameter modifies the character set of the database. This parameter is optional, and the default value is set to AL32UTF8. Please note that, this parameter will set the character set only when a new database is created i.e. a host system directory is mounted using -v option while running the image (Please refer to *Mounting Podman volume/host directory for database persistence* section).
- **ORACLE_PWD:**
  This parameter modifies the password for the `SYS, SYSTEM and PDBADMIN` users. This parameter is optional, and the default value is randomly generated. Note: Using this option exposes the password as a container environment variable. Hence using Podman secrets as described below is recommended.
- **Podman Secrets Support:**
  This option is supported only when **the Podman runtime is being used** to run the Oracle AI Database 26ai Free container image. Podman secrets is a secure mechanism of passing strings of text to the container, such as ssh-keys, or passwords. To specify the password for `SYS, SYSTEM and PDBADMIN` users securely, create a secret named `oracle_pwd` and a secret named `oracle_pwd_priv_key` for the key to encrypt the password. The commands are as follows:
  - Generate the private and public key files

  ```
  mkdir /opt/.secrets/
  ```

```
openssl rsa -in key.pem -out key.pub -pubout
```

- Create a text file with the unencrypted password
Seed password and saving the /opt/.secrets/pwdfile.txt

```
vi /opt/.secrets/pwdfile.txt
```

- Encrypt the password file using the private key generated above

```
openssl pkeyutl -in /opt/.secrets/pwdfile.txt -out /opt/.secrets/pwdfile.enc -pubin -inkey
/opt/.secrets/key.pub -encrypt
rm -rf /opt/.secrets/pwdfile.txt
```

- Create podman secrets for the encrypted passwords and the public key generated above

```
podman secret create oracle_pwd /opt/.secrets/pwdfile.enc
podman secret create oracle_pwd_priv_key /opt/.secrets/key.pub
```

- Pass the podman secrets to the container
Pass the above created podman secrets `oracle_pwd` and `oracle_pwd_priv_key` to the container with the `--secret` option

```
podman run --name <container_name> --secret=oracle_pwd --secret=oracle_pwd_priv_key
container-registry.oracle.com/database/free:latest
```

- **Mounting Podman volume/host directory for database persistence:**
To obtain database persistence, either a named Podman volume or a host system directory can be mounted at the location `/opt/oracle/oradata` inside the container. The difference between these two options are as follows:

  1. If a **Podman volume** is mounted on the `/opt/oracle/oradata` location, then the volume is prepopulated with the data files already present in the image. In this case, the startup will be very fast, similar to starting the image without mount. These data files exist in the image to enable quick startup of the database.

     To use a podman volume for the data volume, run the following:

container-registry.oracle.com/database/free:latest

2. If a **host system directory** is mounted on the `/opt/oracle/oradata` location, two things can happen :
   - If the datafiles are present in the host system directory which is being mounted, then these files will overwrite the files present at `/opt/oracle/oradata` and the database startup will be very fast.
   - If host system directory doesn't contain the datafiles the data files present at `/opt/oracle/oradata` will be overwritten , and **a new database setup will begin**. It takes a significant amount of time (approximately 10 minutes) to set up a fresh database.

   To use a directory on the host system for the data volume, run the following:

   $ podman run -d –name \
   -v :/opt/oracle/oradata \
   container-registry.oracle.com/database/free:latest

Note: Cleaning up a mounted host directory can be done by changing into the directory location and executing `rm -rf * .*`

- **ENABLE_ARCHIVELOG and/or ENABLE_FORCE_LOGGING:**
  These options are enabled by default in the prebuilt database image. To disable these options, execute the relevant SQL commands using SQLPlus from within the database container.

Changing the Default Password for SYS User

On the first startup of the container, a random password is generated for the database if a password is not provided by using the **-e option**, as described in the "*Custom Configurations*" section.

To change the password for these accounts, use the `podman exec` command, to run the `setPassword.sh` script that is found in the container. Note that the container must be running before you run the script.

For example:

```
podman exec <oracle-db> ./setPassword.sh <your_password>
```

Database Alert Logs

```
podman logs <oracle-db>
```

Connecting to the Oracle AI Database Free Container

After the Oracle AI Database indicates that the container has started, and the `STATUS` field shows `(healthy)`, client applications can connect to the database.

Connecting from Within the Container

You can connect to the Oracle AI Database by running a SQL*Plus command from within the container using one of the following commands:

```
podman exec -it <oracle-db> sqlplus sys/<your_password>@FREE as sysdba

podman exec -it <oracle-db> sqlplus system/<your_password>@FREE

podman exec -it <oracle-db> sqlplus pdbadmin/<your_password>@FREEPDB1
```

Connecting from Outside the Container

By default, Oracle AI Database exposes port 1521 for Oracle client connections, using Oracle's SQL*Net protocol. SQL*Plus or any Oracle AI Database client software can be used to connect to the database from outside of the container.

To connect from outside of the container, start the container with the `-P` option, as described in the detailed Podman run command in the "Custom Configurations" section.

Discover the mapped port by running the following command:

```
podman port <oracle-db>
```

To connect from outside of the container using SQL*Plus, run the following commands:

```
# To connect to the database at the CDB$ROOT level as sysdba:
sqlplus sys/<your password="">@//localhost:<port mapped="" to="" 1521="">/FREE as sysdba
```

```
# To connect to the default Pluggable Database (PDB) within the FREE Database:
sqlplus pdbadmin/<your password="">@//localhost:<port mapped="" to="" 1521="">/FREEPDB1
```

Reusing the Existing Database

If the database is started using a host system directory mounted at an `/opt/oracle/oradata` location inside the container, as explained in the Custom Configuration section under "*Mounting Podman volume/host directory for database persistence*", then the data files remain persistent even after the container is destroyed. Another container with the same data files can be started by reusing the host system directory.

To reuse this directory on the host system for the data volume, run the following commands:

```
podman run -d --name <oracle-db> -v \
                     <writable_host_directory_path>:/opt/oracle/oradata \
                     container-registry.oracle.com/database/free:latest
```

Running Scripts After Setup and On Startup

The Container images can be configured to run scripts after setup and on startup. Currently, `.sh` and `.sql` extensions are supported. For post-setup scripts, mount the volume `/opt/oracle/scripts/setup` to include scripts in this directory. For post-startup scripts, mount the volume `/opt/oracle/scripts/startup` to include scripts in this directory.

After the database is set up or started, the scripts in those folders are run against the database in the container. SQL scripts are run as SYSDBA, and shell scripts are run as the current user. To ensure proper order for running scripts, Oracle recommends that you prefix your scripts with a number. For example: `01_users.sql`, `02_permissions.sql`, and so on.

Note: The setup scripts will not run in Free and Free Lite image as both comes with a pre-built database. If host mount point (empty directory) is provided, then a new database setup will start and setup scripts will run.

The following example mounts the local directory `/home/oracle/myScripts` to `/opt/oracle/scripts/startup`, which is then searched for custom startup scripts:

```
container-registry.oracle.com/database/free:latest
```

Oracle True Cache

Oracle True Cache is an in-memory, consistent, and automatically managed cache for Oracle AI Database.

Setting Up Network for Communication Between Primary Database and True Cache Container

- Oracle AI Database Free True Cache container (True Cache container) and the Oracle AI Database Free Primary Database container (Primary Database container) must be on the same podman network to communicate with each other.\
  Set up a podman network for inter-container communication using the following command which creates a bridge connection enabling communication between containers on the same host.

  podman network create tc_net

  Fetch the default subnet assigned to the preceding network by running the following command:

  podman inspect tc_net | grep -iw 'subnet'

  Pick any two IP addresses from the preceding subnet and assign one for the Primary Database container (say, PRI_DB_FREE_IP) and the other for the True Cache container (say, TRU_CC_FREE_IP).

  For communication across hosts, create a macvlan or ipvlan connection per documentation. \
  Specify the preceding podman network using the –net option to the podman run command of both the Primary Database container and the True Cache container as shown in following sections.

Running Oracle AI Database Free True Cache in a Container

- Launch the Oracle AI Database Free Primary Database container using the `podman run` command as follows:

  podman run -td –name pri-db-free \
  –hostname pri-db-free \

```
-p :1521 \
–secret=oracle_pwd \
–secret=oracle_pwd_priv_key \
–add-host=”tru-cc-free:” \
-e ENABLE_ARCHIVELOG=true \
-e ENABLE_FORCE_LOGGING=true \
-v [:]/opt/oracle/oradata \
container-registry.oracle.com/database/free:latest
```

Ensure that your Primary Database container is up and running and in a healthy state.

- Launch the Oracle AI Database Free True Cache container using the `podman run` command as follows:

```
podman run -td –name tru-cc-free \
–hostname tru-cc-free \
–net=tc_net \
–ip \
-p :1521 \
–secret=oracle_pwd \
–secret=oracle_pwd_priv_key \
–add-host=”pri-db-free:” \
-e TRUE_CACHE=true \
-e PRIMARY_DB_CONN_STR=:1521/FREE \
-e
PDB_TC_SVCS=”FREEPDB1:sales1:sales1_tc;FREEPDB1:sales2:sales2_tc;FREEPDB1:sales3:sales3_tc;FRE
\
-v [:]/opt/oracle/oradata \
container-registry.oracle.com/database/free:latest
```

Note: In above command, the list of Primary and True Cache services are mentioned using the string “FREEPDB1:sales1:sales1_tc;FREEPDB1:sales2:sales2_tc;FREEPDB1:sales3:sales3_tc;FREEPDB1:sales4:sa
The string consists of multiple entries in the format “::” and these entries are separated by “;”.

Using the Lite image

Starting an Oracle AI Database 26ai Free Lite Container

the container, run the following command where,`<oracle-db>` is the name of the container:

```
$ podman run -d --name <oracle-db> container-registry.oracle.com/database/free:latest-lite
```

When the container is started, a random password is generated for the `SYS, SYSTEM, and PDBADMIN` users. This is termed as the default password.

The Oracle AI Database is ready to use when the `STATUS` field shows `(healthy)` in the `podman ps` output.

Custom Configurations

To facilitate custom configurations, the Oracle AI Database container provides configuration parameters that you can use when starting the container. For example, this is the detailed `podman run` command supporting all custom configurations:

```
podman run --name <container name=""> \
-p <host port="">:1521 -p  \
-e ORACLE_PDB=<your pdb="" name=""> \
-e ORACLE_PWD=<your database="" passwords=""> \
-v [<host mount="" point="">:]/opt/oracle/oradata \
container-registry.oracle.com/database/free:latest-lite

Parameters:
   --name:        The name of the container (default: auto generated).
   -p:            The port mapping of the host port to the container port.
                  The following ports is exposed: 1521 (Oracle Listener).

   -e ORACLE_PDB: The Oracle AI Database PDB name that should be used (default: FREEPDB1).
   -e ORACLE_PWD: The Oracle AI Database SYS, SYSTEM and PDB_ADMIN password (default: auto g
   -v /opt/oracle/oradata
                  The data volume to use for the database.
                  Has to be writable by the Unix "oracle" (uid: 54321) user inside the conta
                  If omitted the database will not be persisted over container recreation.
   -v /opt/oracle/scripts/startup
                  Optional: A volume with custom scripts to be run after database startup.
                  For further details see the "Running scripts after setup and on startup" s
```

Invoking APIs to any of these features may result in errors:
- Oracle True Cache
- Oracle XML DB
- Oracle Database Multilingual Engine (MLE)
- Oracle Secure Backup (OSB)
- Oracle Memory Speed (OMS)
- Oracle SQL access to Kafka
- Oracle File Mapping (ORAMAP)
- Online Analytical Processing (OLAP)
- Oracle DBNest
- Oracle Database File System (DBFS)
- Oracle Heterogeneous Services
- Oracle Internet Directory
- Oracle External Job Scheduler
- Non-Volatile Memory (NVM) support
- Oracle Data Pump
- Oracle Summary Advisor
- Oracle Notifications service (ONS)
- Oracle Embedded R Execution
- Oracle I/O Numbers
- PL/SQL Server Pages (PSP)
- PL/SQL Hierarchical Profiler
- PLSQL Native Compilation
- Oracle Text - Knowledge Base Extension Compiler
- Oracle Text - Lexical Compiler
- Oracle Text Lexers - Only BASIC_LEXER Type is supported

Availability/Scalability Features

- Oracle Sharding
- Oracle Recovery Manager (RMAN)

Diagnostic Features

- Autonomous Health Framework (AHF)
- Automatic Diagnostic Repository
- Oracle Trace File Analyzer

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Oracle AI Database 26ai Free Container Image Documentation

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Last updated: 10 October, 2025

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23.26.2.0-lite-arm64	linux/arm64	786.81 MB	docker pull container-registry.oracle.com/database/free:23.26.2.0-lite-arm64	2 months ago	eb56b0029432
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latest	linux/arm64	3.25 GB	docker pull container-registry.oracle.com/database/free:latest	2 months ago	e380148ec436
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23.26.2.0	linux/amd64	3.38 GB	docker pull container-registry.oracle.com/database/free:23.26.2.0	3 months ago	8dfd189f9fd7
23.26.1.0-arm64	linux/arm64	3.48 GB	docker pull container-registry.oracle.com/database/free:23.26.1.0-arm64	5 months ago	b129f74fa818
23.26.1.0	linux/arm64	3.48 GB	docker pull container-registry.oracle.com/database/free:23.26.1.0	5 months ago	b129f74fa818
23.26.1.0-lite-arm64	linux/arm64	776.24 MB	docker pull container-registry.oracle.com/database/free:23.26.1.0-lite-arm64	6 months ago	961db65ca528

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